

An Adaptive IDS Based on CIL for Continuously Evolving Threats

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Intro



Objective

Develop an IDS system to detect attacks on network traffic by using Class Incremental Learning

1. **Datasets** 📁
 2. **Methodology** ⚙️
Strategies (ER, DER, iCaRL) to prevent catastrophic forgetting
 3. **Results** 📈
 4. **Conclusion** 📌
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Datasets



Datasets

We used the following Datasets

UNSW-NB15 (2015)

- 49 flow based network features
- 9 attack categories + normal
- 175,341 training samples
- 82,332 testing samples

It was suitable for multi class intrusion detection.

It was a clean predefined split which made it good for controlled incremental tasks

CICIDS2017 (2017)

- 80+ network flow features
- Real enterprise traffic (5 days capture)
- Multiple modern attacks (DDoS, PortScan, Web, Botnet, Brute Force, etc.)

It was more complex and imbalanced
It had better simulation of real evolving threats



Datasets

Relevance to Our Adaptive IDS

- Both datasets contain multiple attack classes
- Allow simulation of sequential class introduction
- Enable evaluation of catastrophic forgetting
- Support testing of replay strategies:
 - iCaRL
 - Experience Replay
 - DER++

Class Distribution

- Classes are imbalanced
- Some attacks have significantly fewer samples
- Imbalance makes catastrophic forgetting more critical

Methodology

Methodology

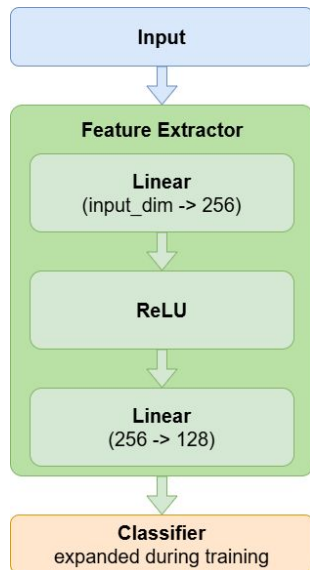
Data preprocessing

Removing
non-informative
features

Removing rows with
missing and infinite
values

Removing
underrepresented
classes

Model definition



Replay strategies

3 memory-based
methods to mitigate
catastrophic forgetting:

- iCaRL
- Experience Replay
- Dark Experience Replay++



Methodology

iCARL

(Incremental Classifier and Representation Learning)

Stores representative exemplars per class (herding selection)

Knowledge distillation on past classes

ER

(Experience Replay)

Stores random past samples in memory

Uses standard cross-entropy loss

DER

(Dark Experience Replay)

DER++ version suitable for highly imbalanced datasets

Replay buffer also stores logits produced by the model

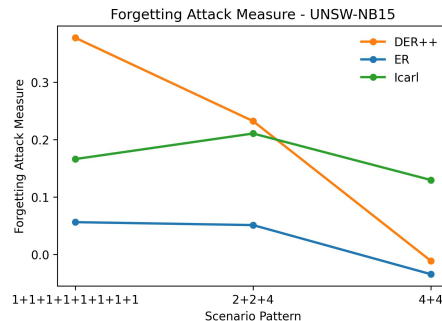
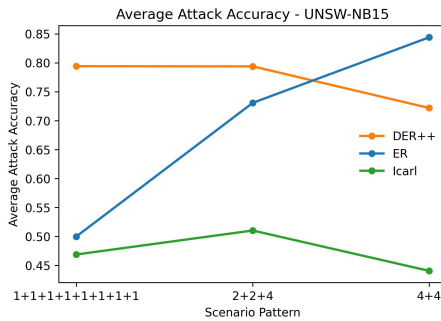
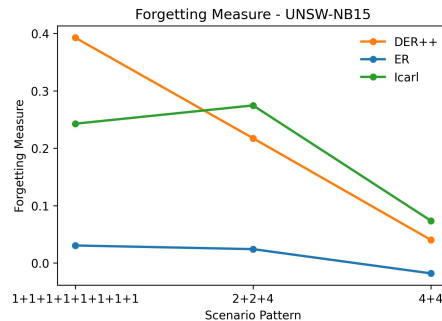
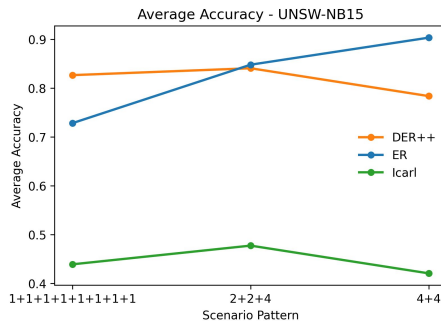
Results



UNSW-NB15 (2015)

Scenario	Metric	iCaRL	DER++	ER
1+1+1+1+1+1+1+1	AA	0.4391	0.8266	0.7282
	AAA	0.4685	0.7940	0.4996
	AF	0.2428	0.3928	0.0306
	AAF	0.1661	0.3771	0.0563
2+2+4	AA	0.4774	0.8407	0.8480
	AAA	0.5100	0.7937	0.7307
	AF	0.2747	0.2175	0.0243
	AAF	0.2106	0.2321	0.0512
4+4	AA	0.4206	0.7837	0.9034
	AAA	0.4402	0.7218	0.8440
	AF	0.0737	0.0403	-0.0180
	AAF	0.1297	-0.0110	-0.0342

Table 2: Results CIL Dataset UNSW-NB15 (2015)

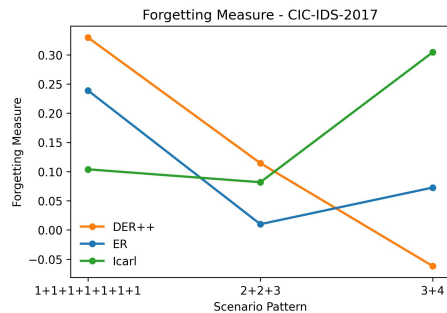
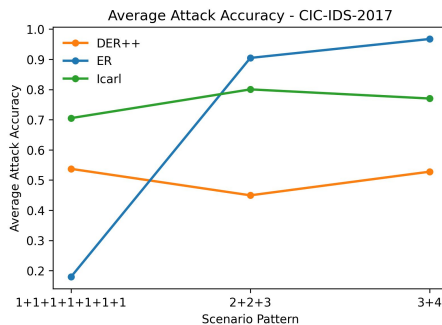
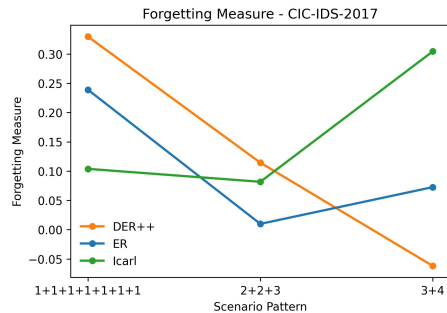
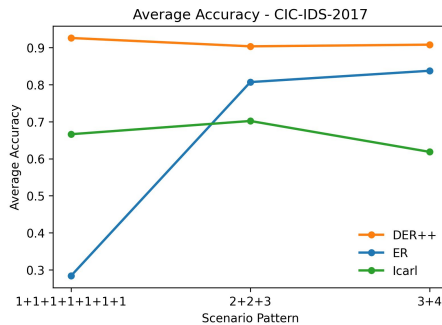




CICIDS2017 (2017)

Scenario	Metric	iCaRL	DER++	ER
1+1+1+1+1+1	AA	0.6662	0.9261	0.2836
	AAA	0.7052	0.5372	0.1794
	AF	0.1039	0.3297	0.2389
	AAF	0.0186	0.0990	0.3597
2+2+3	AA	0.7020	0.9036	0.8070
	AAA	0.8004	0.4495	0.9046
	AF	0.0818	0.1144	0.0099
	AAF	0.0675	-0.1892	-0.0020
3+4	AA	0.6185	0.9081	0.8376
	AAA	0.7702	0.5278	0.9673
	AF	0.3045	-0.0618	0.0727
	AAF	0.0904	-0.3296	-0.0616

Table 1: Results CIL Dataset CICIDS-2017



Conclusion



Conclusion

- **DER++** outperforms ER and iCaRL in complex scenarios (e.g., 1+1+1+1+1+1+1 sequence), offering strong retention of learned attack patterns while adapting to new classes
- Attack category **order** impacts performance but real-world scenarios may lack optimal class ordering